

Chong Tze Zhao

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PROFILE

Proof over promise.

Final-year Computer Science student targeting **quantitative research engineering** / **quant development**. Strong in Python systems, ML/data pipelines, cloud services, and validation-heavy systematic research; selected writeups at chongtzezhao.github.io.

EDUCATION

National University of Singapore Singapore
Bachelor of Computing in Computer Science *August 2023 – May 2027 (Expected)*
• NUS Merit Scholar; GPA: 4.58/5.00

TECHNICAL SKILLS

Languages: Python, Go, C/C++, Java, TypeScript, SQL
ML / Data / Systems: LightGBM, PyTorch, pandas, numpy, numba, PyArrow/Parquet, asyncio/WebSockets, Docker, Linux, SLURM, AWS (ECS/Fargate, S3, Lambda, CloudFormation), React, Express, FastAPI

INDEPENDENT QUANTITATIVE RESEARCH

Self-directed systematic-trading research, Feb–Jun 2026. Short summary only; deeper writeups and artifacts are linked from chongtzezhao.github.io.

Validation-first trading research stack | *Python, LightGBM, numba, PyArrow, SLURM* Feb – Jun 2026
• Built research infrastructure for tick ingestion, market-microstructure features, walk-forward evaluation, slippage modeling, leakage audits, and live/paper execution experiments.
• Published first technical writeup on backtest fill realism: diagnosed a touch-based market-making simulator where roughly **65%** of fills were unsupported by a stricter order-book-delta fill model.
• Built an AFML-style pipeline (CUSUM events, triple-barrier labels, purged/embargoed CV) and used real-fill slippage calibration plus leakage checks to reject strategy variants that did not survive honest costs.

SOFTWARE ENGINEERING EXPERIENCE

Software Engineer Intern Jan 2026 – Jun 2026
Synapse (Singapore's national HealthTech agency) Singapore
• Designed and shipped **two internal monitoring dashboards** covering scheduled batch jobs and Oracle Service Bus integration traffic; deployed to staging on **AWS ECS Fargate** behind an internal ALB.
• Architected a reusable decoupled **producer/consumer pattern** (Python writer → **S3** → Node/Express + React SPA) provisioned via **CloudFormation**, cutting time-to-build for the follow-on dashboard.
• Authored 2,000+ lines across **React 19 / TypeScript, Express 5 / Python**, and CloudFormation IaC, reaching **~90%** test coverage.

Software Engineer Intern Jun 2025 – Aug 2025
Cinch Asia Singapore
• Launched two company-wide AI services in **Python** and **Docker**, reducing manual data processing by **30%**.
• Rewrote 8 legacy **AWS Lambda** scripts into consolidated **Golang microservices**, reducing cold-start latency by **75%** and cutting compute costs by **20%**.
• Implemented CI/CD and monitoring pipelines (**GitHub Actions**, Docker) to automate deployment and improve reliability.

Teaching Assistant | *CS2100 Computer Organisation* Aug 2024 – May 2025
National University of Singapore Singapore
• Tutored 100+ students weekly in data representation, pipelining, memory hierarchy, and assembly; built lab materials on processor architecture and performance optimization.

ACTIVITIES

Quant Engineer (early exposure) Sep 2023 – May 2024
NUS Investment Society — Quantitative Finance Department Singapore
• First exposure to systematic trading: implemented a **mean–CVaR** portfolio optimizer and backtested allocation strategies on historical market data for the department's research track.

SELECTED PROJECTS

BrainHack — TIL AI Competition | *Python, FastAPI, Docker, PyTorch, CUDA* May – Jun 2024
• Co-led a 5-person team integrating **ASR, NLP**, and **VLM** models into an end-to-end multimodal pipeline served via a containerized **FastAPI** inference microservice (sub-100ms responses); **Finalist** among 50+ teams.